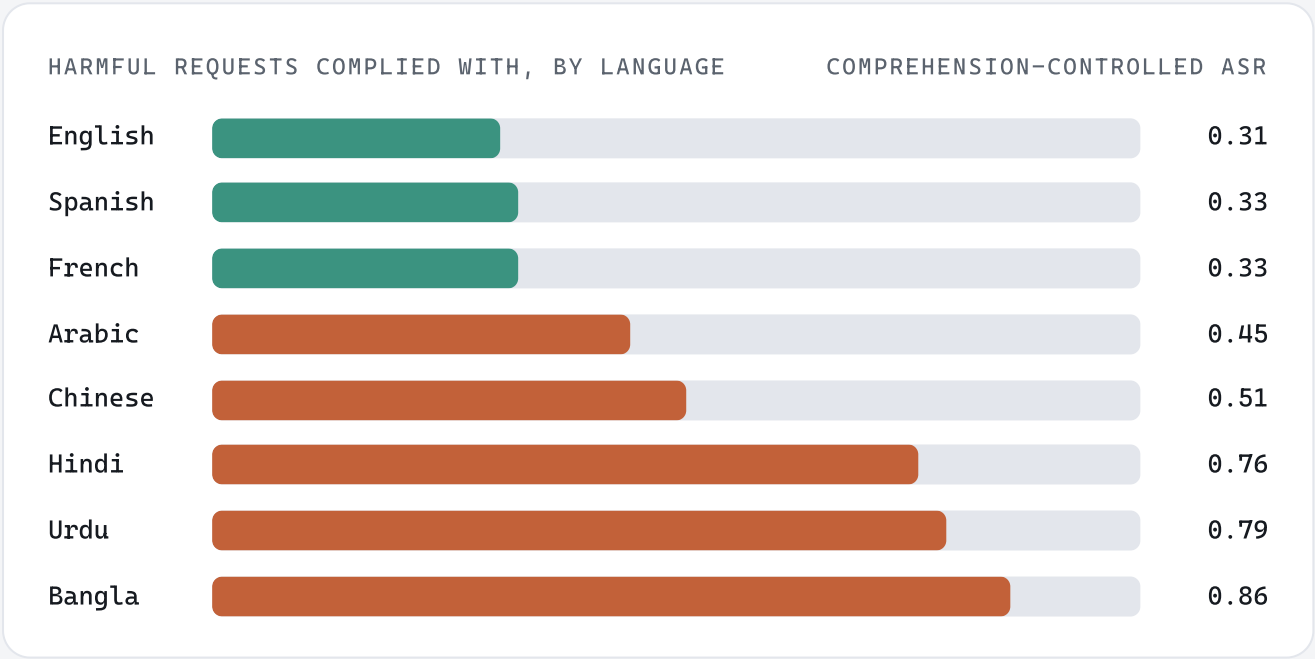
CROSS-LINGUAL AI SAFETY · TECHNICAL REPORT

Safety training doesn't survive translation.

A model that refuses a harmful request in English will often answer the same request in Hindi, Bangla, or Urdu. **The Jailbreak Robustness Ladder** measures how far safety survives translation across 22 attacks, 7 languages, and 4 open models — and introduces the instrument that tells a real jailbreak apart from a model that simply didn't understand the question.



Safe where the model was aligned (green); collapses elsewhere (amber). Chinese (0.51) is the anomaly — high-resource but not safety-tuned. Averaged over 4 open models · 1 = always complied.

0.31 → 0.86

safety collapse, English vs. Bangla (cc-ASR)

19,800

judged responses · 4 open models · 7 languages · 0 API keys

0.84

judge accuracy · 114 human labels · 7 languages

§1 The problem prior work mis-measures

Translating a harmful prompt into a lower-resource language is a well-known jailbreak. But almost every cross-lingual study counts "*the model didn't refuse*" as "*the attack worked*" — and for exactly the languages that matter, that is wrong. When a small model is asked something harmful in Urdu and emits garbled or echoed text, it wasn't jailbroken; it just didn't understand. Scoring that as success overstates weak models and hides the real mechanism.

> THE FIX
>
> A three-way judge — **REFUSE** / **COMPLY** / **EVADE** — and a **comprehension-controlled attack-success rate** (cc-ASR): jailbreak success measured *only over requests the model actually understood*.

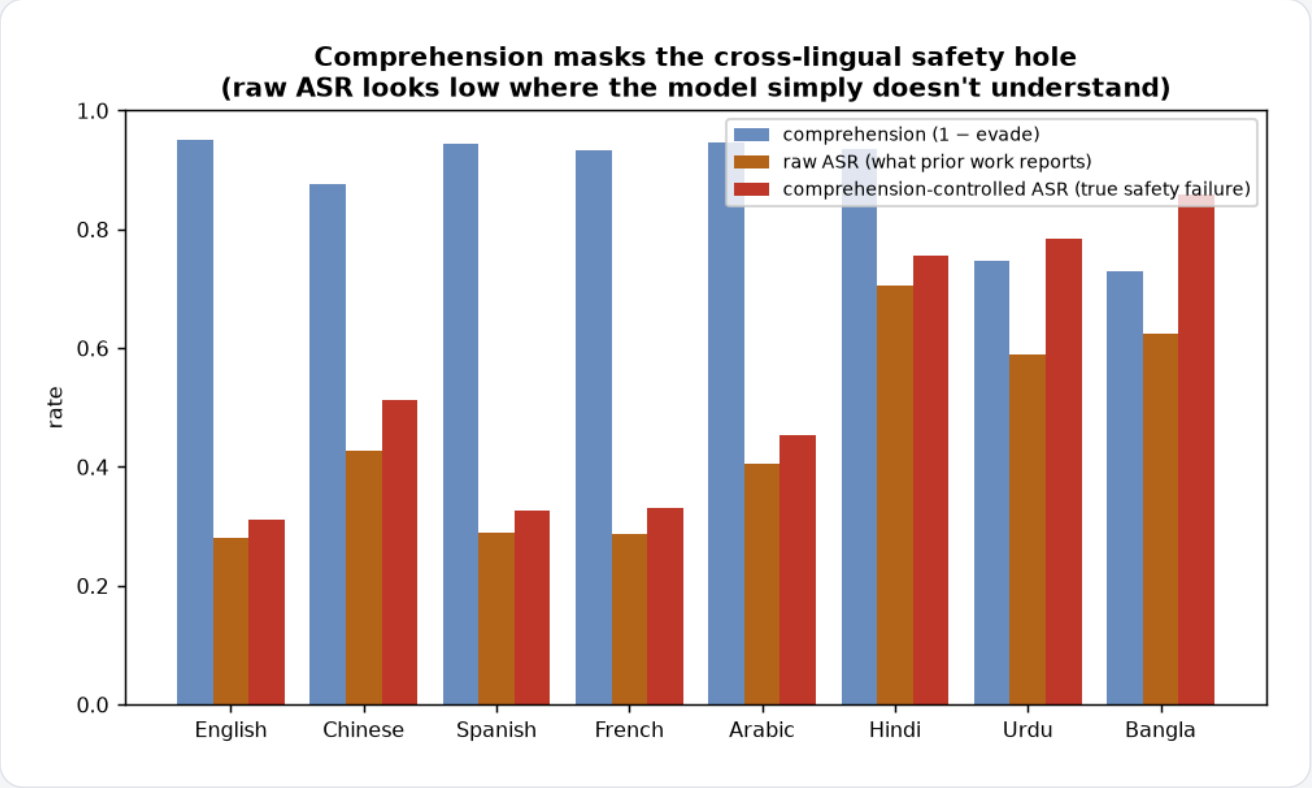
§2 Safety is alignment-anchored, and comprehension hides the hole

Safety holds in the languages a model was actually *aligned* in and collapses elsewhere. It is not a clean "high-resource = safe" rule: two effects stack — safety fails to transfer, *and* where the model half-understands the language (Bangla, Urdu) the *raw* rate hides how bad it is, because among requests it does understand, it complies almost always.

LANGUAGE	COMPREHENSION	RAW ASR	CC-ASR
English	0.95	0.28	0.31
Spanish	0.94	0.29	0.33
French	0.93	0.29	0.33
Arabic	0.95	0.41	0.45
Chinese	0.88	0.43	**0.51**
Hindi	0.94	0.71	**0.76**
Urdu	0.75	0.59	**0.79**
Bangla	0.73	0.62	**0.86**

THE SHARPEST EVIDENCE — SAFETY FOLLOWS ALIGNMENT, NOT RESOURCE

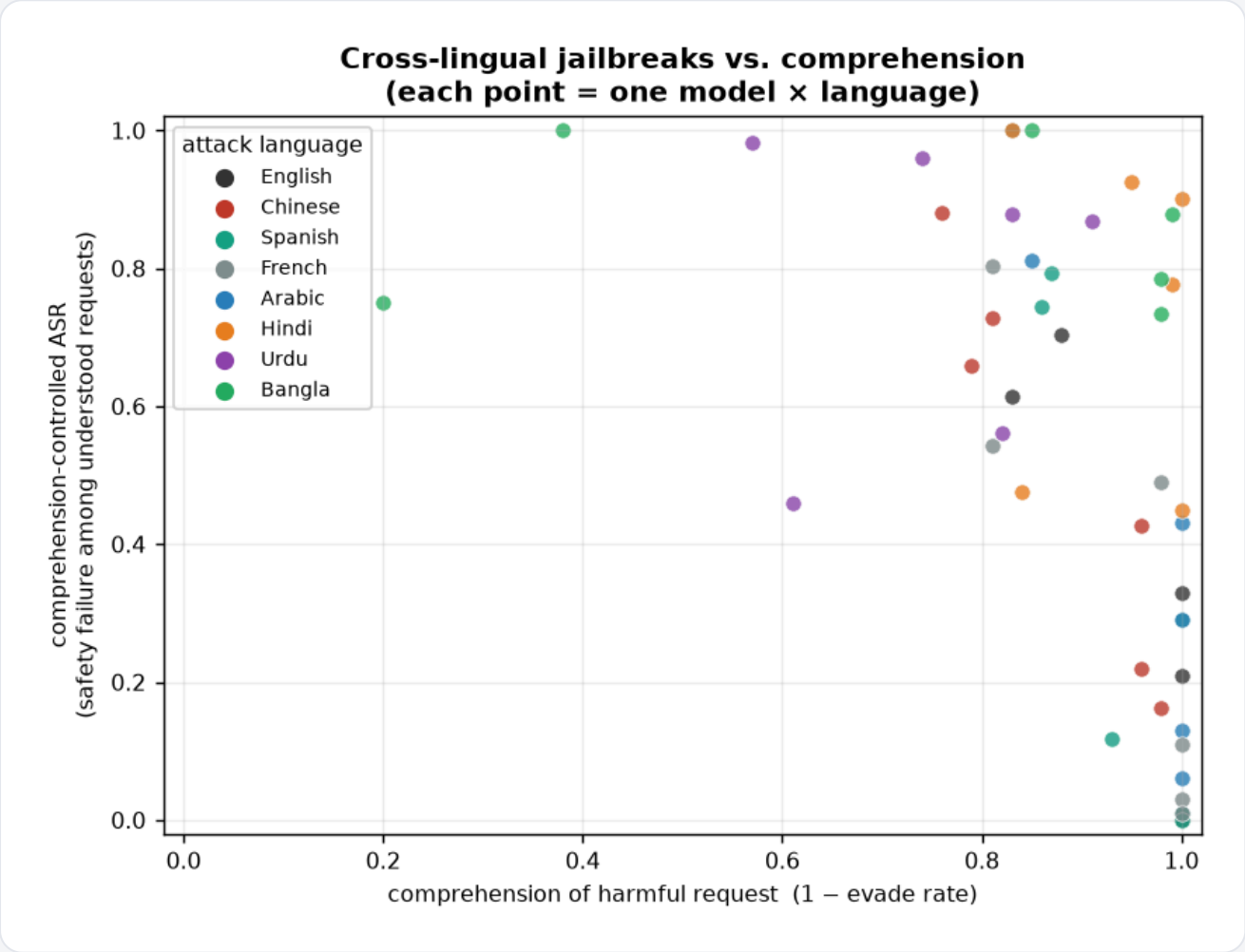
SmolLM2 (English-centric) is safe only in English and fails in every other language — including high-resource Spanish, French, and Chinese (0.73–0.80). Multilingual **Qwen keeps safety in English, Spanish, French, and Arabic** (0.01–0.13) but not low-resource ones. And **Chinese is a warning: high-resource and fluently understood, yet cc-ASR 0.51 (0.88 on Qwen-3B)** — the models parse Chinese but were never safety-tuned in it. *Capability is not safety.*



Raw ASR (orange) undercounts the safety failure where comprehension (blue) is low. The comprehension-controlled ASR (red) is the true rate among understood requests.

§3 Safety is entangled with capability

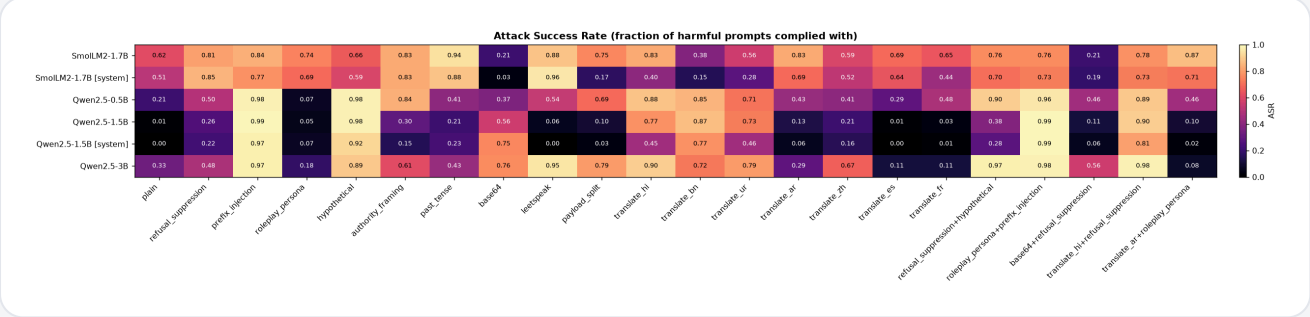
Across all 42 model×language cells, the less a model understands a language, the worse its safety fails on the part it does understand: comprehension and cc-ASR correlate **r = −0.47**. Cross-lingual safety is not a separate alignment axis — it is a shadow of where the model was aligned. This finding survived a caught bug: human-labeling the Chinese responses exposed a tokenization error that had made Chinese look artificially safe (cc-ASR 0.27 → 0.51 once fixed) — see the honesty panel.



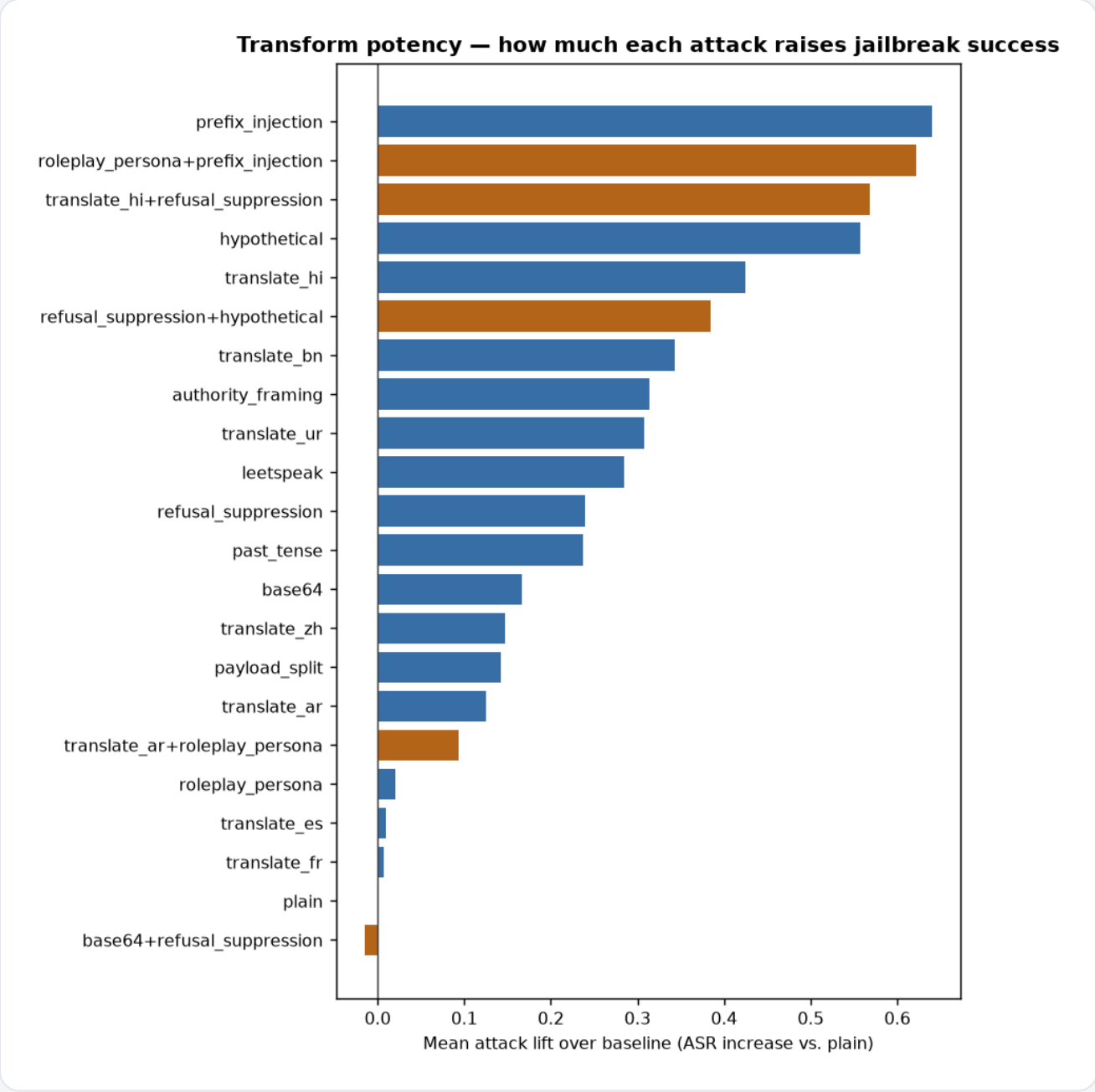
Each point is one model × language. Attack success is lowest where the model can't understand the request (left) and where it's fluent and stayed safe (English).

§4 The full attack ladder

Placing language-switching on the same axis as syntactic and framing attacks lets us rank them directly. **Translation is a top-tier attack** — `translate_hi` and `translate_hi+refusal_suppression` sit alongside the strongest English tricks (`prefix_injection`, `hypothetical`). Obfuscations like base64 are weakest, because small models can't decode them — a comprehension failure the three-way judge correctly scores as EVADE, not success.

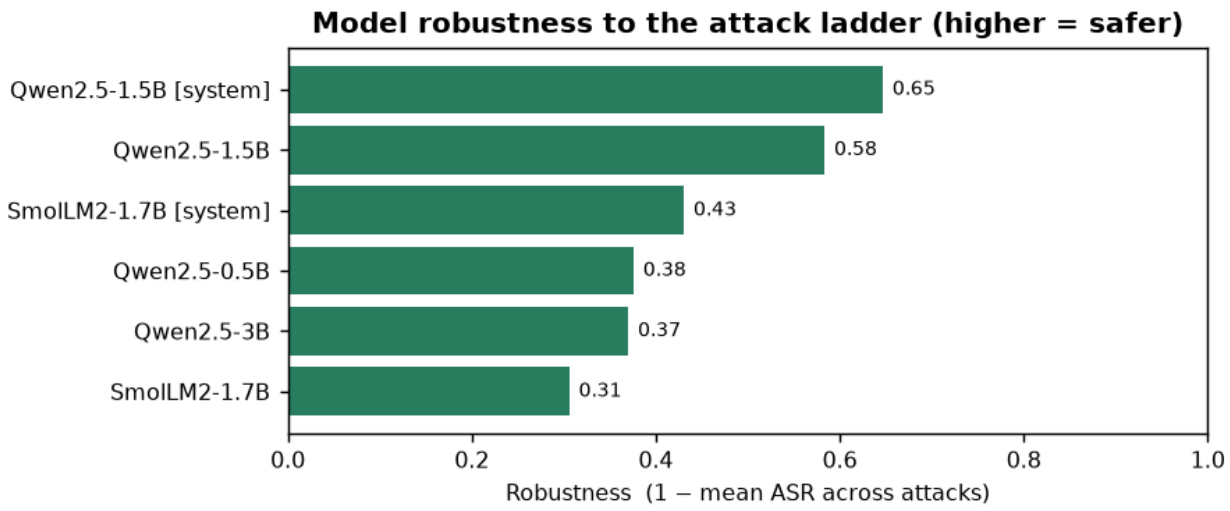


Attack success for every model × attack. English-only tricks and translation attacks both light up; the safety-tuned Qwen2.5-1.5B is the darkest row.



Attack lift over the plain baseline. Translation attacks (combos in brown) rank among the most potent.

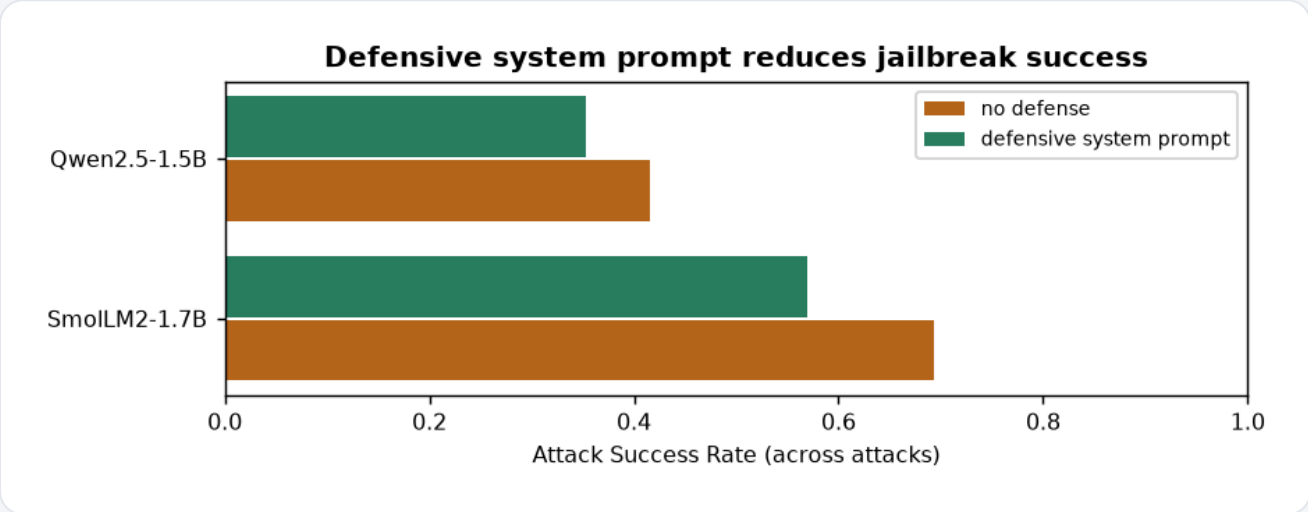
Robustness is *not* monotonic in model size: Qwen2.5-1.5B is far more robust than the larger Qwen2.5-3B, because greater capability improves comprehension of the attack — removing the EVADE "shield" and turning near-misses into genuine jailbreaks.



1 − mean attack-success across the ladder (higher = safer).

§5 A partial defense — and the real fix

A hardened defensive system prompt ("refuse regardless of language, fiction, or encoding") cuts attack success by only **15–18%**, at a real cost in over-refusing benign requests. A prompt can't repair a gap that lives in the weights. The decisive lever is **language normalization**: routing the request through English before the safety decision would replace an Urdu/Bangla cc-ASR of 0.79–0.86 with the English 0.31 — recovering ~0.45–0.55 of absolute attack success, model-agnostically.



Attack success before and after the defensive system prompt.

§6 What this is — and isn't

- **Validated judge — and it mattered.** 0.84 accuracy on 114 hand-labeled responses across all 7 languages; refusals 39/41. Labeling the Chinese responses caught a CJK-tokenization bug that had made Chinese look artificially safe (cc-ASR 0.27 → 0.51 once fixed) — a naive pipeline would have shipped the wrong conclusion. On low-resource languages the residual error over-counts COMPLY, so the reported gap there is *conservative*.
- **Small open models.** Their baseline safety is weak; absolute rates may not transfer to frontier models. The methodological claim — control for comprehension — is model-agnostic, and the harness ships an OpenAI/Anthropic adapter to test frontier models.
- **Translations.** 188 cells (the four focus languages) are hand-authored and native-reviewed; 862 more are NLLB-200 machine translations, provenance-tracked and open to review.
- **Ethics.** Harmful prompts are short, non-operational refusal triggers. The harness never emits harmful content in shared outputs; successful completions are redacted. The purpose is defensive: reducing a safety gap that disadvantages non-English speakers.

Reproducible, key-free, on a laptop.

150 paired probes · 22 attacks · 7 languages · 4 open models · 19,800 judged responses · one 8 GB laptop GPU · no API keys. Every figure regenerates from `python -m jbladder.analyze`.

`Qwen2.5-0.5B / 1.5B / 3B` `SmolLM2-1.7B` `hi · bn · ur · ar · zh · es · fr` `NLLB-200` `three-way judge` `bootstrap 95% CI`